

GUIDELINES to the ESA TENDER CONDITIONS: Additional information related to the tender

VOLUME 1 - Technical Proposal

Due to differences between projects, a standard list of contents cannot be specified. However, while not exhaustive, the following contents are to be considered when preparing the Technical Proposal element of the tender:

- Requirements/specifications:
 - A Specification Tree
 - Specifications
- Design references:
 - Mission assessment
 - System analysis
 - System budgets
 - Concept of designs
 - Mechanical design
 - Thermal design
 - Electrical design
 - Software
 - Interfaces
 - Integration and Test logic
 - Operations logic (e.g. autonomy, availability, FDIR)
- Ground Support Equipment
- Compliance Matrix with Technical System Requirements
- Draft Delivery Lists (Documents, Hardware, and Software)
- Cost drivers:
 - Cost saving options
- Schedule critical developments and contingencies:
 - Schedule critical path
 - Technically complex
 - Technological Readiness Level (TRL) assessment
- New technology items and other risk factors
 - New or innovative technical solutions
 - Technology development
- PA/QA/Safety aspects
 - Single Point failure

Where the ESA Tender Conditions require different content than those mentioned above in the Technical Proposal, these should be provided. Where specific contents are not required, the Tenderer should consider the above list as the basis for his proposal, with additions as he feels necessary.

VOLUME 2 - Management & Administrative Proposal¹

A. The Management Plan

1. Overall Contents

The Management Plan is proposed to describe how the project organisation, including the proposed subcontractors, will operate in conjunction with their company organisations, in order to accomplish the contract objectives, within the framework of the Agency's Management Requirements. The Management Plan may be proposed either as one integrated plan, or as separate dedicated plans for each function. It describes the complete organisation and its management (Project Management, Project Control, Engineering, Product Assurance, etc.) and demonstrates the integration and interaction of these management functions. In addition, it describes the supporting infrastructure that provides the Agency with the necessary information for decision-making and management, that allows efficient management of its own activities and those of lower tier Contractors throughout the Industrial structure, and at the interface to the Agency. The Management plan shall identify the roles and responsibilities of each member in the Industrial Structure.

Following contractor selection, the Management Plan shall be updated and resubmitted for final approval, to reflect the contract negotiations. Specific reference shall be made to each of the following management functions:

- Project Control
- Risk Management
- Project Breakdown Structures
- Schedule Management
- Cost Estimations
- Financial and Cost Management
- Contract Change Management
- Configuration Management
- Document Management
- Logistics and Inventory Management
- Project Phasing and Review Management
- Progress & Performance Evaluation/Management
- Information Management
- Product Assurance

The Management Plan shall document the following areas in detail, in order that the Tenderer and the Agency may agree that all matters are adequately addressed:

¹ Note: Please note that the term "Customer" may refer to the Agency or otherwise the direct customer at lower level of the contractual hierarchy.

Organisation:

Roles and responsibilities
 Authorities of key personnel
 Interfaces between all organisation elements
 Legal agreements and special arrangements between organisation elements

Procedures covering:

Contractor management, including a subcontracting plan describing the policy and the procedures for the selection and management of subcontractors
 Progress reporting
 Cost Control
 Interface management
 Composition and conduct of boards, meetings and reviews

Facilities:

Accommodation and location of project staff
 Computer hardware & software to be deployed
 Communication means

Action Item Control
 Procurement and functional support
 Problem resolution
 Quality assurance
 Engineering
 Safety

- Agency access to data
- Information, data and documentation management and security

2. Describing the Project Management & Control Organisations

A project control organisation is to be established, providing effective management and control of its own activities, and those of its subcontractors, in a proactive manner. The description of the project control organisation is an important element of the Management Plan. Particular attention shall be given to describing not only the internal structure, roles and responsibilities of the project team, but also its external relationships and positioning within the Tenderer's corporate structure.

The following Project Control aspects are to be considered when developing the Management Plan:

- Responsibility for maintenance of all project control data.
- Responsibility to inform the Agency of all critical situations that could lead to adverse programmatic impacts to the project.
- Its role in the implementation of the applicable contract and management requirements.
- Its reporting line to the project manager.
- Its access to the contractor's data as necessary.
- Its place in its project organisation such that fundamental management control functions can be performed effectively, and work performed by others in the organisation can be actively and effectively controlled.
- The support of the Tenderer's management and access to relevant levels of management, both internally and between members of the industrial consortium.

Where a Prime Contractor is proposed, its organisation, roles, responsibilities, authorities and interfaces are to be described in detail, with due consideration of the following aspects:

- Responsibility for the overall implementation and execution of the industrial development programme, including overall management, cost and schedule performance, technical performance, direction of the industrial consortium, and implementation of the Agency's industrial policy requirements.

- The need to establish a project management organisation to act as the single point authority within its company and vis-à-vis the Agency, for the agreement on and execution of the work assigned to its company.
- The nomination of a Project Manager with a team dedicated to the project.
- The responsibility for the project team and their lines of reporting.
- The Project Manager's access to the company management hierarchy for resolution of conflicts.
- The separation of project management for the current project from other projects and operations.
- The need for the Project Management Office to monitor all technical, managerial and administrative activities, provide the directive to accomplish the project, expedite resolution of problems, and interface with the Agency's Project office with identification of Key Personnel.
- Leadership of the Industrial team to ensure compliance with the requirements.

B. Project Requirements Documents for Lower Tier Contractors

To ensure a uniform approach to the applicability of requirements and the performance of project management functions and procedures, throughout the contractual hierarchy, the Tenderer shall propose Project Requirements Documents to be placed on its lower tier Contractors, demonstrating a cost-effective tailoring of requirements to be met during contract execution, in line with the specific project's profile. The tailoring shall be such as to ensure that:

The contractual obligations will be met.

Mandatory requirements specified by the Customer are included.

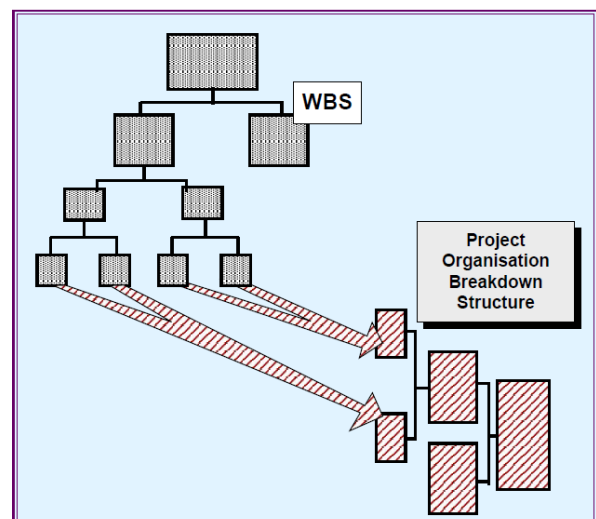
No additional risks are introduced.

No unnecessary requirements are added.

C. The Project Organisation Breakdown Structure

Identification of responsibility for all items of work is essential. This facilitates correction of problems and aids in establishing correct channels of communication. The Project Organisation Breakdown Structure (OBS) depicts the proposed project organisation, as opposed to a Tenderer's Company OBS, which depicts the functional aspects of the company.

It shows the key personnel to be deployed on the work, who will also be identified in the contract, and the assigned responsible parties for each Work Package in the WBS. The following project management functions are to be included as appropriate:



Project Manager
 Contracts Officer
 Project Controller
 Configuration Management Officer
 Documentation Management Officer
 Information Systems Officer
 Product Assurance and Safety Officer
 Technical Officer(s) responsible for design, development, verification of the product(s) under its responsibility

D. The Country/Company Grouping (CCG)

The Agency's obligation to record Industrial Return to the Member States requires identification and recording of expenditures by country. Identifying the contractor for each Work Package allows for the interrogation of data from the contract/Contractor/country view by use of the CCG. The CCG shows the relationships between countries, companies and proposed contracts.

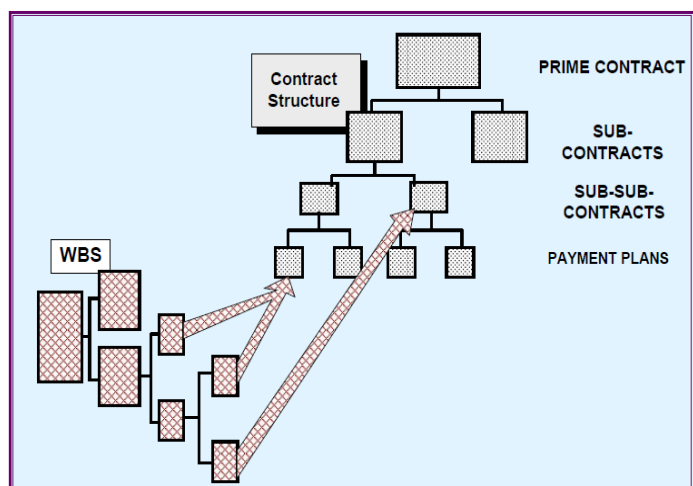
Where practical, such as in projects with simple contractual hierarchies, it may be combined with the Contract Structure, providing meeting Industrial Return policy is not complicated as a result. The Country/Company/Grouping and Contract Structure are typically determined by the Prime Contractor where a complex contracting scheme is envisaged.

The Tenderer develops the CCG by taking into account all work and the proposed contracts/subcontracts it intends to place (Suppliers are included where they affect Industrial Return).

Countries and Contractors shall be identified by the codes, where available, as specified in the current version of ECOS. Contract element coding is to be agreed with the Agency.

E. The Contract Structure

The Contract Structure (CS) depicts the proposed contractual hierarchy in the project. The contract and Contractor codes used shall be the same in both the CCG and CS. The CS for the total Project is to be consistent.



VOLUME 3 - Implementation Proposal

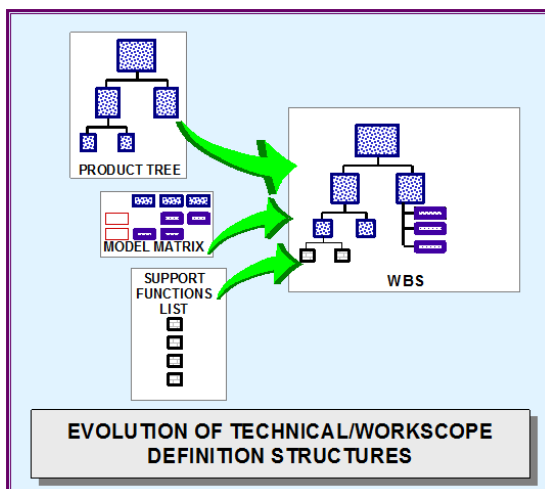
A. Assessment of Major Areas of Risk

To minimise risk, the Tenderer shall state in the proposal any identified risks resulting from any source, including:

- New or innovative technical solutions
- Technology development
- Inadequate, unstable or unclear requirements
- Level of definition and stability of external interfaces
- Adequacy, availability and quality of resources, facilities and tools
- Short-time scales
- High-cost items and work packages
- Industrial organisation

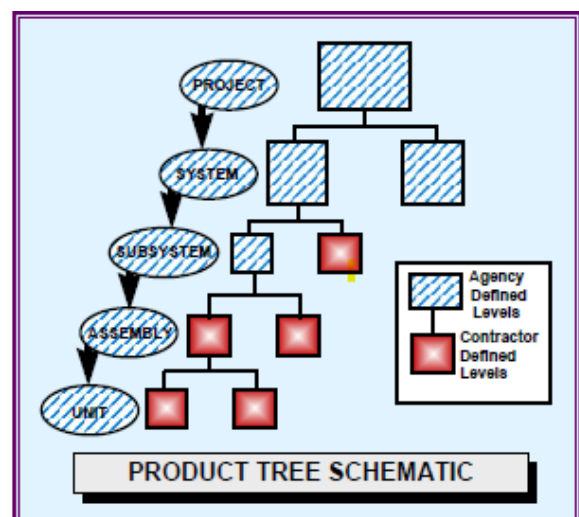
Tenders will be assessed to verify that the level of risk is adequately identified and assessable and that the risk is adequately covered (See Annex 6 “Identifying the Areas of Risk”) and acceptable.

The Product Tree



The following structures are developed in an evolutionary manner, as the project progresses through the definition steps. As such, they reflect the increasing knowledge of the system requirements and design and require the minimum effort to produce.

The Product Tree (PT) establishes a uniform agreed definition of the project's constituent systems, subsystems, assemblies, units, etc., for use by all project participants, and to provide a basis for the development of the WBS. The Product Tree depicts a product-oriented breakdown of the project into successive levels of detail, down to the configuration items. Based on any input given by the Agency, the Product Tree is further developed by the Tenderer proposing to perform the work, with the approval of their customers. The breakdown should be



limited to items where management control is required for any of the following aspects:

Engineering
Configuration Management
Product Assurance
Schedule
Cost
Operations and Logistics – all items that are the subject of a delivery are to be included.

The Product Tree **excludes**:

Supporting functional work, such as Project Management
The development models, such as Qualification Model
as these are components of the WBS.

The Product Tree for the total Project is to be consistent and shall incorporate the Product Trees of proposed lower tier Contractors.

The Model Matrix

The Model Matrix describes the applicability of models for each element of the Product Tree, and their utilisation. It ensures the model philosophy, identified in the Development Plan, is realised. It also provides a means to support the development of the WBS from the Product Tree.

The Work Breakdown Structure

1. Objectives of the WBS

The Work Breakdown Structure (WBS) is the primary project management structure. It is produced by extending the Product Tree elements to include the Agency defined support functions necessary to produce the deliverable end items, the necessary services, and the appropriate development models, by reference to the Model Matrix as applicable. It is used as:

- A common focal point for data creation, collection, summarisation and reporting, and performance monitoring in, and between, all other functions of project management such as Schedule Management, Cost Management, etc. for all project participants.
- A means to define deliverable end items and tasks to be performed.
- A basis for tender comparisons and contract negotiations.
- A means to support the definition of Control Work Packages for management reporting and visibility.
- A basis for the definition of technical interfaces and other relationships between project elements and participants.
- A basis for the development of the Documentation Tree and relating specifications to end items.

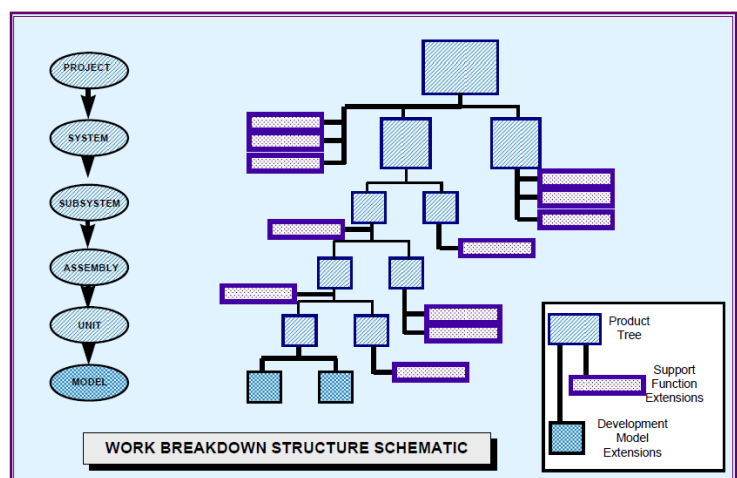
The development of the Work Breakdown Structure begins during the tendering process to:

Obtain a systematic description of the tasks to be performed.
 Provide a basis for evaluation of the work content.
 Provide a framework for the management of any subsequent contracts.

Thus, it is important that the terminology used throughout the tender is consistent with the Tender Action.

A top-down approach is employed to the development of the WBS to ensure all major work areas are considered and to avoid omissions and duplications. Starting with the Product Tree as a basis, the Tenderer shall produce its part of the WBS by:

- Reviewing each element of its Product Tree.
- Appending the required development models as children to the applicable elements.
- Adding the required support functions as children of the element to establish the Work Packages.
- Determining Work Packages on the basis of management complexity.
- Incorporating the WBS's of its proposed contractors
- Confirming, by means of a bottom-up approach, the completeness of the WBS:
- All models and work items are included.
- The correct location of each Work Package beneath its parent product.
- The correct inclusion of all products, assemblies, subsystems and systems.



To ensure work is included at the appropriate level, work related to manufacturing, assembly, integration and test shall be shown against the models, and Engineering work shall be identifiable in connection with its related Product Tree element, such as System, Subsystem or Unit level.

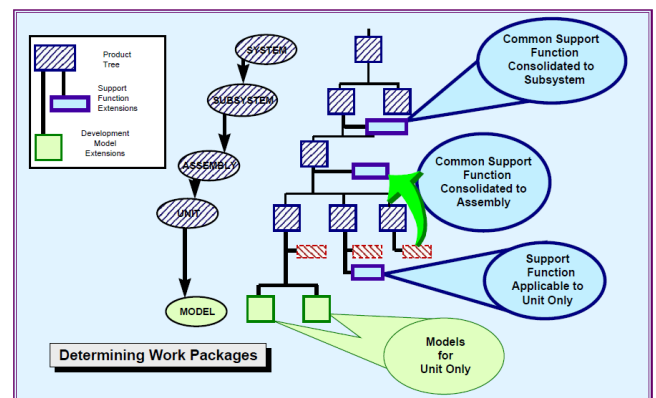
Establishing Work Packages in the WBS

Work Packages are established primarily for the purposes of establishing cost, planning and other baselines, and for the collection of progress information, such as actual costs and man-hours expended. They are used in these processes by the Contractor responsible for the work, and as such are determined at a level of detail adequate to allow control to be exercised by the Contractor. This requires that work packages be defined in sufficient detail to relate each with:

The lowest level contractor applicable.
 The most detailed, required schedule data.
 The most detailed, required cost data.
 The most detailed, required contractual deliveries.

While reporting to ESA may be at a summary level (See Annex 5 - “Determining Control Work Packages” below), data collection shall be performed at the Work Package level, and thus allow more detailed reporting and data submission when requested by ESA.

Support functions may be shown in the WBS at any level, but effort must not be duplicated. Support Functions used are those defined in the Support Function File of the current version of the Agency's costing system, ECOS. This supports a common definition both within the project and between projects, aiding the collection, comparison, and understanding of data. ECOS is available to all Tenderer on a free of charge basis. The Tenderer shall use the contents of the Support Function File for reference.



In addition to product related Work Packages, Work Packages, which are functional in nature can be one of two types:

- Apportioned Effort (i.e. effort related in direct proportion to product-oriented Work Packages).
- Level of Effort (i.e. effort which cannot be directly associated with any end products but only with the total project or contract).

Each Work Package shall:

- Have a unique identifier and concise description, and a clearly defined start and end.
- Wherever possible, produce deliverable items and be confined to one development model. For control purposes, especially Schedule and Cost Management, non-product related work is difficult to assess in terms of performance and value to the project. Where such work is required, it should, wherever possible, be apportioned to product-oriented work for measurement and control purposes.
- Be established on the basis of management complexity:
 Be manageable by one person.
 Contain an adequate number of work elements to facilitate control.
 Facilitate the incorporation of contract changes.

Factors to be considered in the final, detailed partitioning of Work Packages are summarised in the table below.

FACTOR	PRODUCT WP's	FUNCTIONAL WP's
Number of End Items (Product complexity)	X	
Models or Production Volume	X	
Apportioned Efforts		X
Functional Specialisations		X
Duration		X
Geographical Locations	X	X
Responsibilities	X	X
Organisations	X	X
Contract Type	X	X
Cost	X	X
Activities (Project Complexity)	X	X

Work Package Descriptions

Work Package Descriptions shall be completed for each Work Package, to indicate a good and complete understanding of the work involved and the products to be produced.

Contractually, the Statement of Work defines the work to be performed.

The completion of Work Package Descriptions is intended to demonstrate that consideration has been given to all aspects, to avoid duplication and omissions, thus benefiting both Tenderer and Agency.

The required data shall provide visibility and understanding of the distribution of tasks.

The Tenderer shall use the PSS-A-20 form:

- The form at its latest version is directly downloadable from <https://esastar-publication.sso.esa.int/supportingDocumentation/filter/all> where it can be found under:
 - Supporting Documentation
- PSS Forms

Work Package Description – PSS-A-20

The purpose of this form is to obtain a detailed description of the work to be performed under this Work Package (WP).

Unless specified otherwise in the ESA Tender Conditions, WP descriptions are to be supplied at the task/equipment level. Each form shall contain the following information:

- Name of project.
- Designation of project phase (if applicable).
- Work package reference defined in Work Breakdown Structure.
- Title of Work Package.
- Economic Operator responsible.
- Number of the sheet and total number of sheets pertaining to a given Work Package Description (use continuation sheets as appropriate).
- Major constituent to which the work package belongs. This shall not be limited to

hardware subsystems but shall cover, as appropriate, functional task areas, such as management, product assurance, etc.

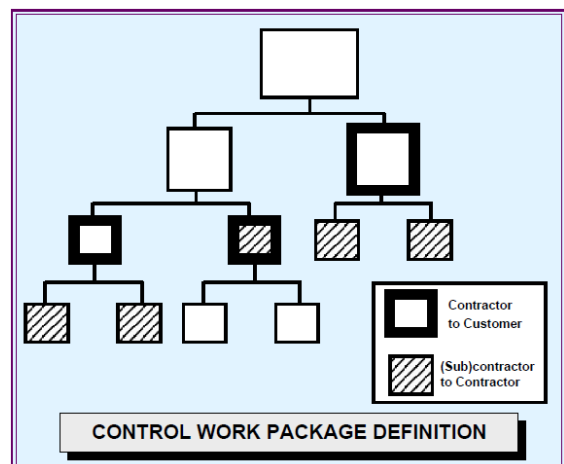
- Identifier of the start event which denotes commencement of the work package activities on the respective network.
- Plan date of the work package start event. The date shall be based on planning calculations.
- Issue number of the Work Package Description.
- Identifier of the end event, which denotes completion of the work package activities on the respective network.
- Plan date of the work package end event. The date shall be based on planning calculations.
- Date of latest issue of the Work Package Description.
- The name of the contractor's employee responsible for the work package (WP manager).
- List of items (documentation and/or hardware) which must be completed, or be available to start the work package.
- List of tasks specifically excluded, if appropriate, in order to increase clarity of the scope of the work package.

Determining Control Work Packages

Control Work Packages are those elements of the WBS for which data and reports, including financial, cost and manpower, are required to be submitted to the Agency.

Control Work Packages:

- Are agreed between the Contractor and its direct Customer during contract negotiations.
- Are at any level of the WBS.
- For reporting purposes, represent the total scope of work of the Contractor.
- May be at different levels of the WBS dependent on the Contractor's level in the Contract Structure.
- Are determined based on the level of visibility of the Contractor's data, required by the direct Customer.



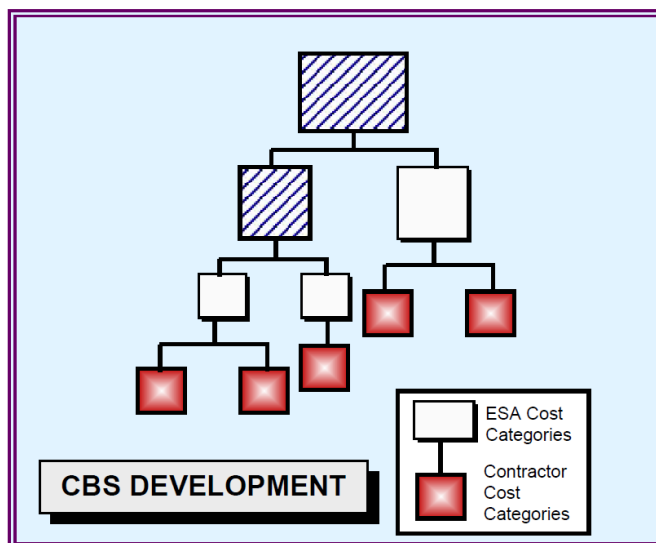
As noted above and depicted in the figure, the selection of Control Work Packages is dependent on the Contractor's level in the contractual hierarchy. The Control Work packages established for a Subcontractor are agreed on the basis of the visibility of its work required by its higher-level Contractor.

In turn, the Contractor's Control Work Packages are established and agreed, with its higher-level Customer, on the basis of the visibility of its work required by the Customer. Normally, these are at a higher level in the WBS, but may be the same as some Subcontractor's Control Work Packages, where the work is of an important or critical nature, and visibility is required to be maintained.

The Cost Breakdown Structure

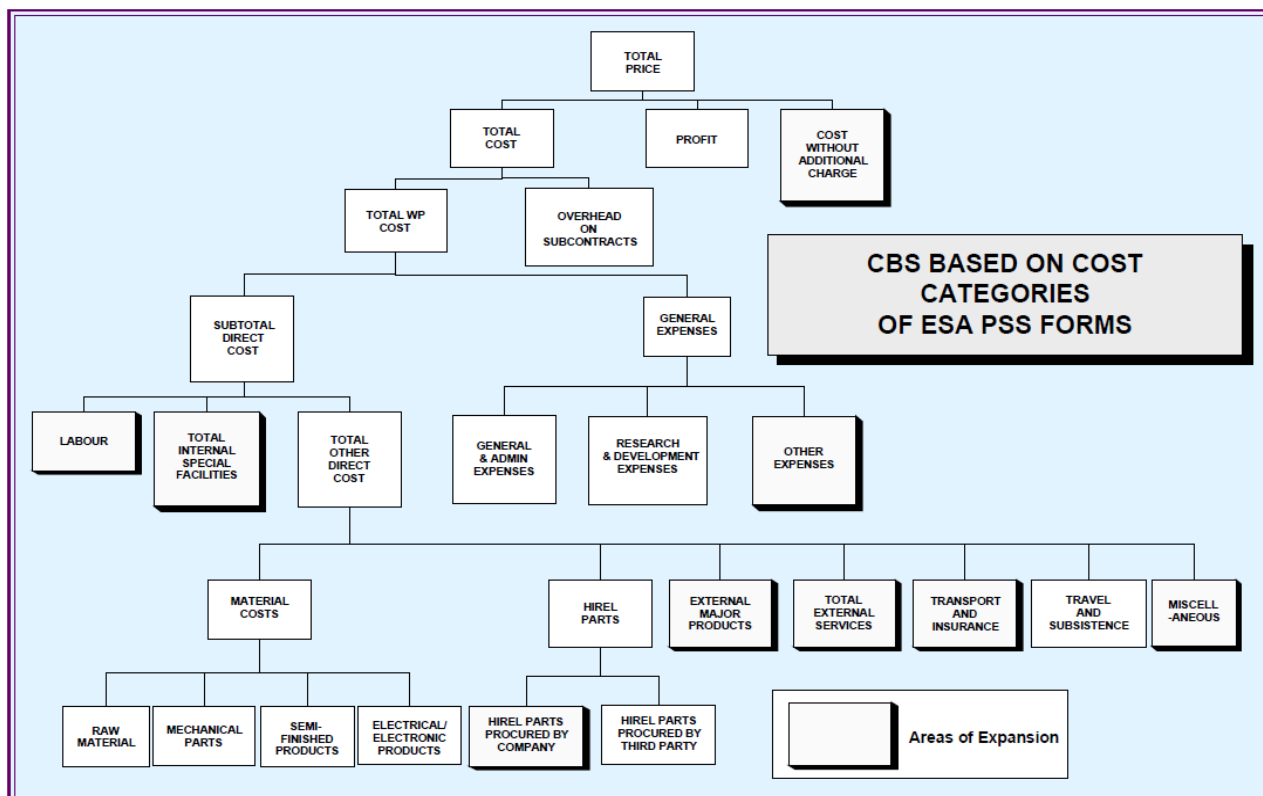
The Cost Breakdown Structure (CBS) defines the breakdown of all the costs of the project, including labour, equipment, facilities, etc., into agreed cost categories to be used for Financial and Cost Management, and provides the framework for cost summarisation with other structures.

To ensure that cost collection and analysis can be performed in the Agency's standard cost categories across the whole project, the CBS is based on the Agency's standard cost headings, as defined in its PSS series of Cost Breakdown Forms.



Where the Agency's cost categories alone are not sufficient, they may be extended to include the Tenderer's own categories, within each applicable Agency category by assigning its internal cost categories to the applicable Agency cost categories, taking account of all work to be performed and resources required. This includes the possibility of assigning only one Contractor cost category to an Agency category, as illustrated above.

The CBS based on the Agency's standard cost headings is shown below, with typical areas of expansion highlighted.



Schedule Requirements

1. Purpose & Objectives

The purposes of proposing schedules are:

- To present the main objectives, in the form of milestones, of the envisaged work.
- To identify the tasks required to achieve these objectives.
- To clearly and realistically present the logical sequence of the proposed tasks.
- To provide the basis for subsequent schedule management, control and reporting during execution of the work.

Basis of Proposed Schedules

The proposed schedules shall take account of, and be based on:

- The schedule and financial targets.
- Detailed analysis of the tasks.
- Critical review of all activity interdependencies and durations.
- Significant interfaces and constraints.
- Available resources.
- Adequate lead times for procurement.
- Availability of planned facilities.
- Normal downtime of facilities.
- Allowances for resolving interferences and incompatibilities during integration and testing.
- Time phasing consistent with cost and manpower data in the Work Package Manpower and Cost Plan, and the Price Projection.

a) Milestones in the Schedule

*i) **Milestones Specified by the Agency***

The milestones specified by the Agency for setting targets for major events and for monitoring schedule progress are to be included in the planning and supporting schedules, developed and presented by the Tenderer.

*ii) **Milestones Specified by the Contractor***

In turn, the Tenderer similarly identifies additional milestones for its work and its lower tier tenderers work, representing key points in the project, which are to be used for target setting and schedule progress monitoring. From these milestones, the Tenderer shall propose Payment Milestones, which when approved by the Customer, shall be separately identified and documented as part of Milestone Payment Plans.

b) Detailed Schedules

Detailed Schedules shall be proposed, which support and include the milestone targets and expand upon the work depicted by the Work Breakdown Structure. A Detailed Schedule shall be produced for each major system constituent and participant. If more than one proposed Contractor is working within a major system constituent, then a Detailed Schedule for each shall be developed, and combined (using the same principles of integration as the creation of Master Schedules from Detailed Schedules), to make a Detailed Schedule for each major system constituent.

These schedules serve as the basis for schedule management between the Contractor and its lower tier contractors during execution of the work, and shall include the activities and tasks, as appropriate and applicable to the level of the work (product/system/subsystem/unit/etc.), associated with:

Design, Development, Production, Construction
Assembly, Integration
Testing, Acceptance, Delivery
Procurement
Reviews
Software development
Support equipment, materials, supplies & services
Logistics and transport

To facilitate progress measurement, and the creation of the Master Schedule, at least the following type of milestones shall be identified and included in the Detailed Schedules:

Payment milestones
System and Subsystem reviews
Software reviews
Interfaces*
Deliveries (Major documents, hardware & software)
Any other as required by the Customer

*Interface milestones shall be defined that represent the logical relationships between proposed Contractors, such as deliveries of items and the completion of required information such as designs, specifications, etc. Each Milestone is included in the schedules of both the provider and the recipient of the item or information, and connected via relationships.

Risks to the attainment of any milestone shall be highlighted, and alternative solutions provided to minimise these. Reference is to be made to the corresponding technical, schedule and cost sections of the tender.

Where risks to a milestone have been recorded, detailed schedules are to be prepared to allow, during execution of the work, early identification of adverse impacts and the implementation of contingency plans where required.

The Detailed Schedules represent the work scope as identified by the WBS. In order to ensure that this is achieved, the Tenderer shall:

- Identify the parent work package for each activity, which may be a Control Work Package or, as is more often the case, a lower level work package.
- Agree these relationships with work performers.
- Delete activities without parent work packages or add work packages if the work is required.
- Create activities for work packages without activities or delete work packages without activities (unless LOE work package).
- Ensure that all tasks in WPD's are included in a Detailed Schedule.

c) Master Schedules

Master Schedules shall be created, by each proposed Contractor, by integrating, at the Control Level, the Detailed Schedules and milestones, for its own work (typically product/system level activities) and that of its proposed lower tier Contractors (typically subsystem/unit level activities).

During execution of the work, following contract negotiations and award, these schedules shall become the basis for routine reporting, and monitoring, progress assessment and coordination by the Agency.

i) The Control Level

In order to provide visibility and traceability of schedule data at the level of detail required by the Agency, the Tenderer shall propose the Control Level (the content and level of detail of the Master Schedule), to be regularly reported and delivered to the Agency. The Control Level for schedule management, and its corresponding Master Schedule is not to be created by simply summarising the Detailed Schedules to their Control Work Package parents, which are primarily determined for Cost Management and other management purposes.

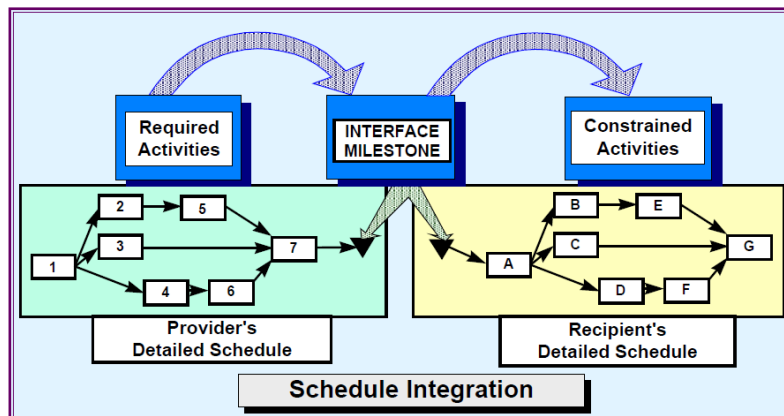
Rather, normally, the Control Level, and its corresponding Master Schedule, are to comprise specified milestones, and the major activities supporting and connecting them. This shall include all Agency specified milestones and other milestones as agreed.

Dependent on the content and level of detail of the proposed Control Level, this may involve the integration of any of the following:

Complete Detailed Schedules (such as product/system level work).
Summaries of the Detailed Schedules (such as subsystem/unit level work).
Specific extracts of the Detailed Schedules.

The Master Schedule may be expanded, during execution of the work, to include more detailed information if circumstances necessitate, for example, where delays are forecast, or where the Agency is required to perform a coordinating role with participants other than industry contractors.

ii) Developing the Master Schedules



To ensure that integrated schedules allow for a complete critical path analysis, the interface milestones, in the Detailed Schedule of the provider, are constrained by all required activities that must be completed in order for the milestone to be achieved.

Similarly, in the Detailed Schedule of the recipient, the interface milestones constrain all activities that are dependent on the completion of the milestone.

After the Detailed Schedules are produced, each proposed Contractor shall:

- Integrate its and all its proposed Contractors Detailed Schedules into a single Master Schedule at the agreed Control Level, including interface milestones between them, and submit it to the Agency.
- Ensure that the work being performed and the Detailed Schedules are in agreement.
- Assess the realism of Detailed Schedules and assess the associated risks.

Project Reviews

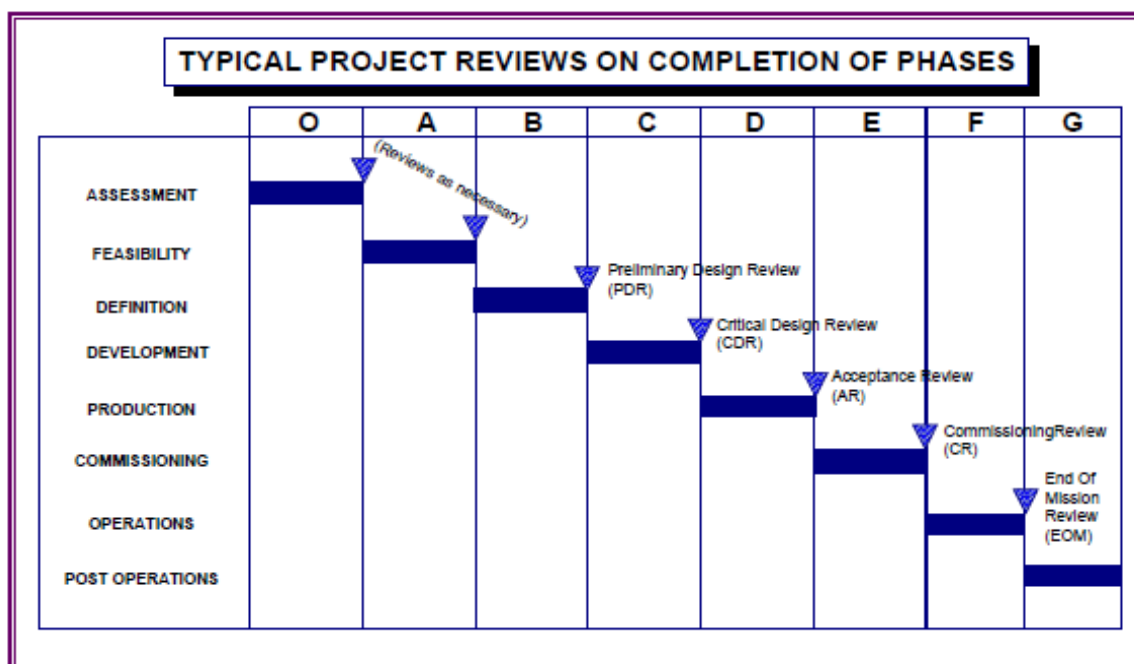
1) The Project Lifecycle

A typical Agency project is comprised of a series of activities performed in a number of phases, punctuated by Project Reviews to assess the status, conformance and progress of the work, and approve the baseline for the next stage of work.

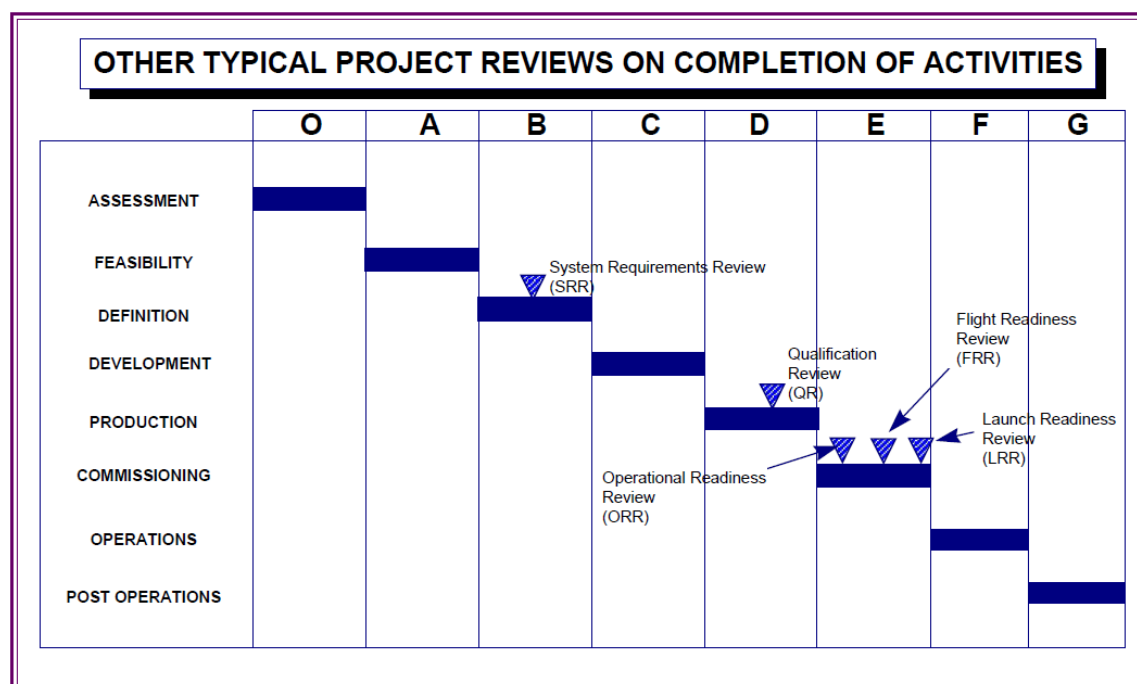
2) Reviews

Each phase is concluded by a Project Review to establish the satisfactory completion of the preceding phase and establish an agreed baseline for the succeeding phase. The typical reviews are illustrated below.

(Reviews as necessary)



Additional reviews are also held at the completion of a major development stage, to assess the development status, such as those illustrated below.



The Agency will, based upon consideration of the size and complexity of the programme concerned, select from the reviews at the programme or project/element level during Phases B to G, those to be considered as Agency reviews.

Reviews applicable to a specific project or element will normally be selected from these reviews. The Contractor for a project/element concerned will propose the reviews to be conducted at lower tiers, and their schedule, ensuring that all technical data required to support a review at a given level (element, subsystem, ...) are properly reviewed and

confirmed at the subordinate levels, in concert with reviews at higher levels. The Agency may participate in any such lower-level reviews.

Reviews are normally organised and chaired by the Customer (by the Agency for Agency reviews). The Agency will chair reviews at the Prime Contractor level and selected Agency reviews at lower levels as noted above.

A number of independent experts will be involved in each review, depending on the project complexity, and the innovative character of the technology and techniques applied. These independent experts are not members of the project team directly involved in the project concerned.

Reviews are normally organised into discipline-oriented review panels, which may include programmatic aspects (cost, schedule, organisation, ...) and conducted in a formal manner with review results being recorded and followed up in a controlled manner.